A $\mathbb{Z}_{37}$ orbit-matrix reduction and cyclotomic obstructions for the conference graph of $H(668)$

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Abstract

A Hadamard matrix of order $668 = 4 \cdot 167$ is the smallest order whose existence is unknown; equivalently no symmetric conference matrix $C(334)$, regular two-graph on 334 points, or strongly regular graph $\mathrm{srg}(333, 166, 82, 83)$ is known. Assuming a fixed-point-free automorphism of order 37, permitted by the eigenvalue-field obstruction (unlike the group-developed Cayley case), we reduce $\mathrm{srg}(333, 166, 82, 83)$ to a 9×9 array of circulants over $\mathbb{Z}_{37}$, the strong-regularity identity becoming a per-character condition $M_t^2 + M_t - 83I = 0$ over $\mathbb{Z}[\zeta_{37}]$. We then (i) completely enumerate, via a sound symmetry reduction, the 2901 admissible 9×9 orbit matrices, the necessary skeletons of any such graph; (ii) prove by two lift-free arguments that *no* cyclotomically-structured (Paley-type) graph exists for the 6 proper nontrivial multiplier subgroups of order ≥ 4; (iii) reduce the two finest multipliers $|H| \in \{2, 3\}$ to a per-skeleton lift we leave open, and prove a *collapse theorem* explaining the difficulty (the negation involution forces every trace-linear invariant, spectral or character-theoretic or association-scheme/Krein, to be determined by the orbit matrix, so any obstruction must be non-linear in the $\{0, 1\}$ connection-set indicators), while confining a hypothetical order-2 solution to θ-multiplicity $a \in \{4, 5\}$; and (iv) document that a base-point gauge-fixing routinely used to accelerate SAT search becomes *unsound* once a multiplier constraint is imposed. Parallel-tempering evidence points to nonexistence. We release validated open-source code and the enumeration; an appendix records a negative result on square-divisor spectral filters for the equivalent Legendre pair LP(333).

1 Introduction

A *Hadamard matrix* of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ and $HH^\top = nI$. The Hadamard conjecture asserts that such a matrix exists for every $n \equiv 0 \pmod 4$. The smallest order for which existence is open is $n = 668 = 4 \cdot 167$ [1, 2]. Several classical objects are each equivalent to $H(668)$; we work with two.

Legendre pairs. For $x \in \{\pm 1\}^\ell$ let $\mathrm{PAF}_x(k) = \sum_{i \in \mathbb{Z}_\ell} x_i x_{i+k}$ be the periodic autocorrelation. A *Legendre pair* (LP) of length ℓ is a pair $u, v \in \{\pm 1\}^\ell$ with $\mathrm{PAF}_u(k) + \mathrm{PAF}_v(k) = -2$ for all $k \not\equiv 0$. An LP of length ℓ yields a Hadamard matrix of order $2\ell + 2$ via the two-circulant-core construction [3]. For $H(668)$ one needs $\ell = 333 = 9 \cdot 37$; LP(333) is open [4, 5].

Conference graphs. A symmetric conference matrix C of order $m = 334$ ($C = C^\top$, zero diagonal, ± 1 off-diagonal, $C^2 = (m-1)I$) yields $H(668)$ via $H = \left[\begin{smallmatrix} C+I & C-I \\ C-I & -C-I \end{smallmatrix}\right]$, and $C(334)$ is equivalent to a regular two-graph on 334 points, whose descendant is the strongly regular graph $\Gamma = \mathrm{srg}(333, 166, 82, 83)$ [7, 2]. The parameter set $(333, 166, 82, 83)$ is listed as open in Brouwer's tables, and no catalogued conference-matrix construction (Paley; Mathon's $q^2(q+2)+1$ family [8]) reaches order 334.

Prior obstructions. It is known that no *group-developed* (Cayley) conference graph on 333 vertices exists: its nontrivial eigenvalues are $(-1 \pm \sqrt{333})/2$ with $\sqrt{333} = 3\sqrt{37} \in \mathbb{Q}(\sqrt{37})$ of conductor 37, while every group of order $333 = 3^2 \cdot 37$ has commutator subgroup inside the Sylow-37 subgroup and hence admits a character of order 3; a Cayley eigenvalue $\chi(S)$ of order-3 character lies in $\mathbb{Z}[\zeta_3] \not\supseteq \mathbb{Q}(\sqrt{37})$, a contradiction (Lemma 1). Likewise the relevant prime 2 is not self-conjugate modulo 167 ($\mathrm{ord}_{167}(2) = 83$), so the Turyn–Schmidt field-descent bounds do not forbid $H(668)$; the barrier is constructive, not arithmetic.

Contributions and related work. We (i) give a $\mathbb{Z}_{37}$ orbit-matrix reduction of Γ permitted by the eigenvalue-field obstruction, distinct from the group-developed case (Section 2); (ii) completely enumerate the 2901 admissible orbit matrices and prove, by two lift-free arguments, the nonexistence of cyclotomically-structured graphs for the 6 multiplier subgroups of order ≥ 4 (Section 3); (iii) for the two finest multipliers $|H| \in \{2, 3\}$, prove a trace-linearity collapse theorem and confine a hypothetical order-2 solution, while documenting a soundness pitfall in gauge-fixed SAT search (Section 3); and (iv) record, in Appendix A, that square-divisor spectral filters do not strengthen the Kotsireas–Koutschan filter for the equivalent LP(333). The orbit-matrix method for strongly regular graphs with a prescribed semiregular automorphism is due to Behbahani–Lam [10] and was developed for two-graphs and self-orthogonal codes by Crnković–Maksimović–Rodrigues–Rukavina [11] and, for regular two-graphs on up to 50 points, by Maksimović [12]. Our contribution is its application to this open conference parameter set: the resulting per-character condition $M_t^2 + M_t - 83I = 0$, the complete skeleton enumeration, the cyclotomic nonexistence theorems, and the trace-linearity collapse that delimits the method.

2 A $\mathbb{Z}_{37}$ orbit-matrix reduction of $\mathrm{srg}(333, 166, 82, 83)$

Lemma 1. *No group of order* 333 *admits a* $(333, 166, 82, 83)$ *partial difference set; equivalently there is no group-developed conference graph on* 333 *vertices.*

Sketch. As in the introduction: a 1-dimensional character of order 3 would force a nontrivial eigenvalue $(-1 \pm 3\sqrt{37})/2 \in \mathbb{Z}[\zeta_3]$, impossible since $\mathbb{Q}(\sqrt{37}) \not\subseteq \mathbb{Q}(\zeta_3)$. □

Lemma 1 rules out the *regular* automorphism group; it does not rule out individual automorphisms. We therefore consider a fixed-point-free automorphism σ of order 37. Since $37 \mid 333$ ($333 = 9 \cdot 37$), such σ partitions the vertices into 9 orbits of size 37; we identify each orbit with $\mathbb{Z}_{37}$ so that σ acts as $+1$. The adjacency matrix becomes a 9×9 array of circulants: writing $D[i][j] \subseteq \mathbb{Z}_{37}$ for the connection set of block (i, j), $A_{(i,a),(j,b)} = 1 \iff (b - a) \bmod 37 \in D[i][j]$, with $D[j][i] = -D[i][j]$ (symmetry) and $0 \notin D[i][i] = -D[i][i]$ (no loops). Crucially the eigenvalue-field obstruction of Lemma 1 does *not* apply: $\sqrt{37} \in \mathbb{Q}(\zeta_{37})$ (the quadratic Gauss sum), so order 37 is permitted. The reduction of a strongly regular graph to circulant blocks under a semiregular automorphism is the standard orbit-matrix method [10, 11]; our contribution is its application to this open parameter set, the resulting per-character condition, and the (non)existence results below.

Proposition 1 (Character reduction)**.** *Define, for $t \in \mathbb{Z}_{37}$, the 9×9 matrix $M_t[i][j] = \sum_{d\in D[i][j]} \zeta_{37}^{td}$. Then A is $\mathrm{srg}(333,166,82,83)$ (equivalently $A^2 + A - 83I = 83J$) if and only if*

$$(t=0) \quad R := M_0 \text{ satisfies } R^2 + R - 83I_9 = 3071\,J_9, \qquad (t \neq 0) \quad M_t^2 + M_t - 83I_9 = 0,$$

where each M_t ($t \neq 0$) is Hermitian over $\mathbb{Z}[\zeta_{37}]$ with spectrum in $\{(-1 \pm \sqrt{333})/2\}$.

The integer matrix $R = (|D[i][j]|)$, the *orbit matrix*, is symmetric with row sums 166, even diagonal, and (from $R^2 + R - 83I = 3071J$) eigenvalues 166 (once) and $(-1 \pm \sqrt{333})/2$ (four times each); in particular $\operatorname{tr} R = 162$ and a Cauchy–Schwarz argument confines the diagonal entries to $\{10, 12, \ldots, 26\}$. We verified Proposition 1 by reconstructing the Paley conference graphs $P(9), P(25), P(49)$ in block-circulant form, confirming agreement of the direct and per-character tests.

3 Computational attack and results

Orbit-matrix census. A backtracking search with knapsack-tight pruning, made complete by a sound symmetry reduction (within each maximal equal-diagonal group of orbits, sort by connection to a fixed orbit), enumerates *all* admissible orbit matrices up to orbit relabeling: exactly 2901 representatives, with every diagonal class exhausted. (An earlier run without the reduction had confirmed ≥ 1678 before several highly symmetric diagonals exceeded a 4×10^9-node cap.) These 2901 form the complete set of necessary skeletons, so the lift-free arguments of Theorem 1 apply to all of them; the remaining obstruction for the fine multipliers is the *lift* to connection sets satisfying Proposition 1 (Remark 1). A representative skeleton (row sums 166, tr = 162, spectrum $\{166,\, r^4,\, s^4\}$) is

$$R = \begin{pmatrix} 10 & 19 & 19 & 19 & 19 & 20 & 20 & 20 & 20 \\ 19 & 14 & 22 & 22 & 20 & 15 & 15 & 18 & 21 \\ 19 & 22 & 14 & 22 & 20 & 15 & 18 & 21 & 15 \\ 19 & 22 & 22 & 14 & 20 & 15 & 21 & 15 & 18 \\ 19 & 20 & 20 & 20 & 18 & 24 & 15 & 15 & 15 \\ 20 & 15 & 15 & 15 & 24 & 20 & 19 & 19 & 19 \\ 20 & 15 & 18 & 21 & 15 & 19 & 24 & 17 & 17 \\ 20 & 18 & 21 & 15 & 15 & 19 & 17 & 24 & 17 \\ 20 & 21 & 15 & 18 & 15 & 19 & 17 & 17 & 24 \end{pmatrix}, \qquad R^2 + R - 83I_9 = 3071\,J_9.$$

Exact lift (SAT). The lift is equivalent to the convolution system, for all $i, j \in \mathbb{Z}_9$, $g \in \mathbb{Z}_{37}$,

$$\sum_{i'} \big(D[i][i'] * D[i'][j]\big)(g) = \mu + (k-\mu)[i{=}j, g{=}0] + (\lambda - \mu)[g \in D[i][j]],$$

which we encode in a Boolean SAT/CP model [9] (membership variables, cardinality, symmetry). For the *general* (unstructured) lift a base-point gauge-fixing removes the 37^8 per-orbit symmetry, and the encoding so validated reconstructs $P(9), P(25)$. We caution that this gauge-fixing is *unsound* once a multiplier (cyclotomic) constraint is added (Remark 1); the cyclotomic results of Theorem 1 therefore rely only on the gauge-free arguments (a),(b).

Call a $\mathbb{Z}_{37}$-symmetric $\mathrm{srg}(333,166,82,83)$ *H-cyclotomic* if every connection set is a union of cosets of a multiplier subgroup $H \leq \mathbb{Z}_{37}^*$ (the Paley-type structure). The next theorem rules these out completely for the 6 proper nontrivial subgroups of order ≥ 4, by two lift-free arguments (no base-point gauge); the two finest cases $|H| \in \{2,3\}$ reduce to a cyclotomic lift that we leave open (Remark 1).

Theorem 1 (No cyclotomically-structured lift, coarse multipliers)**.** *No H-cyclotomic* $\mathrm{srg}(333,166,82,83)$ *exists for $|H| \in \{4, 6, 9, 12, 18, 36\}$.*

(a) *(Via the orbit matrix.) An H-cyclotomic graph forces its 9×9 orbit matrix to have H-coset-compatible cardinalities: off-diagonal $R[i][j] \equiv 0$ or $1 \pmod{|H|}$ (the residue records whether $0 \in D[i][j]$), and diagonal $R[i][i] \equiv 0 \pmod{|H|}$ if $-1 \in H$ else $\pmod{2|H|}$. No integer matrix meets these together with the orbit-matrix identity $R^2 + R - 83I = 3071J$: an independent exhaustive backtracking enumeration over the coset-compatible alphabet finds none for each of the six orders (a complete-search certificate trusting no solver; each search is exhausted in well under a second). Hence no such graph exists, for any orbit matrix.*

(b) *(Independent confirmation, $|H| \in \{12, 18, 36\}$.) The stronger orbit-matrix-free system (every block H-invariant, the graph 166-regular, R unconstrained) is directly* UNSAT.

Both arguments are lift-free, hence independent of any base-point gauge. They were validated by reconstructing the Paley graphs $P(9), P(25), P(49)$ (orbit-matrix-free) and $P(13), P(17)$ (orbit matrix).

Remark 1 (The fine multipliers $|H| \in \{2, 3\}$ are open)**.** The orbit-matrix argument (a) cannot reach $|H| \in \{2, 3\}$: order 2 imposes no congruence (every R is compatible), and the order-3-compatible orbit matrices *do* exist (a complete backtracking enumeration finds exactly 960, all with diagonal $(12, 12, 18, 18, 18, 18, 18, 24, 24)$). Both cases reduce to the per-orbit-matrix cyclotomic lift over the complete set of 2901 admissible orbit matrices. We do not resolve this lift. A base-point gauge-fixing (forcing $0 \in D[0][i]$) that greatly accelerates the *general* lift is *unsound* under a multiplier constraint: a multiplier-invariant set shifted by $c \neq 0$ is no longer invariant (as $(\mathrm{mult} - 1)c = 0 \Rightarrow c = 0$), so the cyclotomic ansatz has no base-point freedom, and forcing $0 \in D[0][i]$ wrongly rejects even the genuine negation-invariant Paley graph $P(25)$. Without that gauge, neither the element-level nor a coset-level SAT encoding decides a single $|H| \in \{2, 3\}$ instance within 30 minutes. We therefore leave $|H| \in \{2, 3\}$ open. The parallel-tempering evidence below points to nonexistence, and for $|H| = 2$ Proposition 2 confines a hypothetical solution to θ-multiplicity $a \in \{4, 5\}$ and the discussion shows it resists every trace-linear obstruction; but a proof remains open.

Unstructured lift (parallel tempering). For the general (unstructured) lift we ran a batched GPU parallel-tempering search minimizing $\sum_{t \neq 0} \|M_t^2 + M_t - 83I\|_F^2$, which is 0 iff the lift succeeds. The search is validated: on the $\mathrm{srg}(25, 12, 5, 6)$ skeleton it drives the objective to 0 (a verified lift) within 18 steps. On $\mathrm{srg}(333, 166, 82, 83)$, across ≈ 450 of the 634 skeletons with 8192 chains and replica exchange per skeleton ($\approx 3.7 \times 10^6$ chains total), the objective plateaued at $\approx 3.9 \times 10^5$ with no lift found.

The order-2 residue: a sharper ceiling. Of the two open cases $|H| \in \{2, 3\}$ (Remark 1), the order-2 (negation-invariant) case admits a sharp structural analysis, recorded here: how far it can be pushed, and why it resists. Since -1 is a quadratic residue mod 37, negation-invariance of every block makes each character matrix M_t a *real* symmetric 9×9 matrix with eigenvalues in $\{\theta, \tau\}$, $\theta, \tau = (-1 \pm 3\sqrt{37})/2$; write a_t for the θ-multiplicity of M_t. The Galois group $\mathbb{Z}_{37}^*/\{\pm 1\}$ permutes the M_t and acts on $\sqrt{37}$ through the Legendre symbol, forcing $a_t = a$ on the quadratic residues and $9 - a$ on the non-residues for a single integer a.

Proposition 2 (Order-2 multiplicity reduction)**.** *For a negation-invariant $\mathbb{Z}_{37}$-symmetric* $\mathrm{srg}(333, 166, 82, 83)$*, the common residue-character θ-multiplicity satisfies $a \in \{4, 5\}$.*

Proof sketch. Expanding $\mathrm{tr}\, M_1 = a\theta + (9 - a)\tau$ in the integral basis $\eta_g = \zeta_{37}^g + \zeta_{37}^{-g}$ via the quadratic Gauss sum $\sqrt{37} = \sum_{g=1}^{18} \left(\frac{g}{37}\right) \eta_g$ identifies the diagonal pair-counts $N_g := \#\{i : \{g, -g\} \subseteq D[i][i]\}$

as $N_g = 3a - 9$ on the nine residue pairs and $N_g = 18 - 3a$ on the nine non-residue pairs. Non-negativity $0 \le N_g \le 9$ gives $a \in \{3, 4, 5, 6\}$. For $a \in \{3, 6\}$ the counts are extremal $\{0, 9\}$, forcing every diagonal block to equal $Q^{\mp}$ (all non-residues, resp. all residues) and hence the orbit matrix to have constant diagonal $(18, \ldots, 18)$; an exhaustive backtracking search, complete under the S_9 orbit-relabeling symmetry (1.9×10^8 nodes, corroborated to 9×10^{10} nodes by an independent reduction), finds *no* admissible orbit matrix with constant diagonal 18. Hence $a \in \{4, 5\}$. The N_g identity was validated on the negation-invariant graphs $P(13)$ and the 25-orbit Paley graph of $\mathrm{GF}(5^3)$. $\square$

That the order-2 case resists our tools is structural, not accidental: the standard linear-algebraic and character-theoretic obstructions are all *trace-linear*, and they collapse.

Theorem 2 (Trace-linearity collapse)**.** *Let A be the adjacency matrix of an order-2-cyclotomic $\mathbb{Z}_{37}$-symmetric* $\mathrm{srg}(333, 166, 82, 83)$ *with orbit matrix R, and let $\varphi \in \{0, 1\}^{333\times 333}$ be the permutation matrix of the negation involution $(i, g) \mapsto (i, -g)$. Then:*

- *(i) $\varphi^2 = I$, $\varphi A = A\varphi$ (so φ is an automorphism of A), and $\varphi J = J\varphi = J$;*
- *(ii) the unital algebra $\mathcal{A} = \langle A, \varphi, J\rangle \subseteq M_{333}(\mathbb{Q})$ is spanned by $\{I, A, J, \varphi, \varphi A\}$, so $\dim \mathcal{A} \le 5$;*
- *(iii) the trace of every element of $\mathcal{A}$ is a $\mathbb{Q}$-linear combination of $\operatorname{tr} I = 333$, $\operatorname{tr} A = 0$, $\operatorname{tr} J = 333$, $\operatorname{tr} \varphi = 9$, and $\operatorname{tr}(\varphi A) = \operatorname{tr} R = 162$, all fixed by the parameters and R.*

Consequently every spectral, character-theoretic, involution-graded, or quotient/Krein invariant of the graph, each of which is the trace of a word in A, φ, J, is determined by R alone. No such invariant can distinguish a hypothetical order-2-cyclotomic solution from the orbit-matrix data, hence none can obstruct existence: any obstruction must be non-linear in the $\{0, 1\}$ connection-set indicators.

Proof. (i) Each block $D[i][j]$ is negation-invariant, so $A_{(i,-g),(j,-h)} = [\,(-h) - (-g) \in D[i][j]\,] = [\,h - g \in -D[i][j]\,] = [\,h - g \in D[i][j]\,] = A_{(i,g),(j,h)}$; thus $\varphi A \varphi^{-1} = A$, and $\varphi = \varphi^{-1}$ gives $\varphi A = A\varphi$. Each row of φ has a single 1, so $\varphi J = J = J\varphi$. (ii) From $A^2 + A - 83I = 83J$ we get $A^2 = 83I - A + 83J$, and 166-regularity gives $AJ = JA = 166J$; by induction $A^n \in \operatorname{span}\{I, A, J\}$ for all n. Since $\varphi^2 = I$ and φ commutes with A and J (with $\varphi J = J$), every word in A, φ, J equals $A^n \varphi^{\varepsilon}$ for some $n \ge 0, \varepsilon \in \{0, 1\}$, which lies in $\operatorname{span}\{I, A, J\}\cdot\{I, \varphi\} = \operatorname{span}\{I, A, J, \varphi, A\varphi\}$ (using $J\varphi = J$); and $A\varphi = \varphi A$. (iii) $\operatorname{tr} A = 0$ (no loops); $\operatorname{tr} \varphi$ counts fixed points (i, g) with $g \equiv -g$, i.e. $g = 0$, one per orbit, so 9; and $\operatorname{tr}(\varphi A) = \sum_{i,g} A_{(i,-g),(i,g)} = \sum_{i,g} [\,2g \in D[i][i]\,] = \sum_i |D[i][i]| = \operatorname{tr} R = 162$, since $g \mapsto 2g$ permutes $\mathbb{Z}_{37}$. Linearity of the trace finishes the claim. $\square$

Remark 2 (Why character and multiplier methods are inert)**.** Two consequences sharpen Theorem 2. (a) In the group ring $\mathbb{Z}[\mathbb{Z}_{37}]$ (generator X) the strong-regularity identity reads $B^2 + B - 83I_9 = 83\,T\,E_9$ with $B = (B_{ij})$ the block-indicator matrix, E_9 all-ones, and $T = \sum_g X^g$ *supported on the principal character*; so under any nonprincipal character $X \mapsto \zeta_{37}^t$ the inhomogeneous term vanishes and one is left with the homogeneous $M_t^2 + M_t - 83I = 0$, whose Gauss-/Jacobi-sum evaluations reduce to cardinalities already fixed by R. (b) The numerical multiplier $83 \equiv 9 \pmod{37}$ has *odd* order $\operatorname{ord}_{37}(83) = 9$ with $-1 \notin \langle 83\rangle$, so 83 is not self-conjugate mod 37 and the Turyn–Schmidt field-descent / multiplier bounds impose nothing, the same arithmetic as the Paley graph $P(13)$, which exists. In particular the moment hierarchy collapses: as A satisfies a quadratic, $\operatorname{tr}(A^k)$ is an affine function of $\operatorname{tr} A$ for every k, so no power-sum (eigenvalue-multiplicity) count beyond R is available.

Thus the order-2 obstruction, if any, lives entirely in the non-linear $\{0, 1\}$-realizability of the connection sets: the lift of Remark 1, which we leave open.

4 Discussion

We distinguish what is proven from what is evidence. *Proven:* Proposition 3 (square-divisor filters do not help LP(333)); Proposition 1 (the reduction); the complete enumeration of the 2901 admissible orbit matrices; Theorem 1 (no cyclotomically-structured $\mathbb{Z}_{37}$ lift for the 6 multiplier subgroups of order ≥ 4, by two lift-free arguments); and Proposition 2 (a hypothetical order-2 solution is confined to θ-multiplicity $a \in \{4, 5\}$ and resists every trace-linear obstruction). *Open:* the cyclotomic lift for the two finest multipliers $|H| \in \{2, 3\}$ (Remark 1), which sound SAT does not decide in feasible time. *Evidence:* the parallel-tempering plateau is a strong indication, from a search demonstrably able to find lifts when they exist, that no unstructured $\mathbb{Z}_{37}$-symmetric lift exists either, though it is not a proof. Taken together with Lemma 1, this makes a $\mathbb{Z}_{37}$-symmetric srg$(333, 166, 82, 83)$ unlikely, and focuses any future construction of $H(668)$ on (a) genuinely asymmetric conference graphs, (b) automorphisms of other orders (for which the orbit-matrix block size grows), or (c) the Legendre-pair / cocyclic routes [13], where Appendix A clarifies which spectral tools do and do not help.

Reproducibility

All code (the $\mathbb{Z}_{37}$ reduction and verifier, the complete orbit-matrix enumerator with its sound symmetry reduction, the orbit-matrix coset-compatibility and orbit-matrix-free cyclotomic checks, the exact and coset-level SAT lifts together with the validation exhibiting the multiplier gauge-fixing as unsound, the GPU batched parallel-tempering search, and the conference-to-Hadamard emitter), together with its validation (reconstruction of the Paley graphs $P(9), P(25), P(49)$, the conference-to-Hadamard chain on $H(12), H(28), H(36)$, and recovery of a genuine lift on the srg$(25, 12, 5, 6)$ skeleton) is openly available at `https://github.com/maceoCK/h668-z37` and archived on Zenodo (DOI to be assigned on release). The verifiable certificates are included: the 2901 enumerated orbit matrices (each checkable against $R^2 + R - 83I = 3071J$ by the bundled verifier) and the independent coset-compatible enumeration giving count 0 for each $|H| \in \{4, 6, 9, 12, 18, 36\}$ (Theorem 1(a)).

A A square-divisor spectral filter for LP(333) adds no pruning

The Legendre-pair route to $H(668)$ needs an LP(333); we record a self-contained negative result on one natural filter for it, independent of the orbit-matrix development above. Write $\mathrm{PSD}_x(s) = \left|\sum_j x_j \omega^{js}\right|^2$, $\omega = e^{2\pi i/\ell}$. Summing the LP condition gives the flat-spectrum form

$$\mathrm{PSD}_u(s) + \mathrm{PSD}_v(s) = 2\ell + 2 \quad (s \not\equiv 0), \qquad \mathrm{PSD}_u(0) + \mathrm{PSD}_v(0) = 2, \tag{1}$$

so the row sums are ± 1; for $\ell = 333$ the constant is $2\ell + 2 = 668$.

For $d \mid \ell$ the *d-compression* $c_r^{(d)} = \sum_{j \equiv r \ (d)} x_j$ satisfies $\mathrm{PAF}_{c^{(d)}}(k) = \sum_{i \equiv k \ (d)} \mathrm{PAF}_x(i)$ [6], and $\mathrm{PSD}_x(\ell/d)$ depends only on $c^{(d)}$.

Cube root (mod 3). At $s = \ell/3$, write $\gamma = c^{(3)}(u)$, with γ_r odd (a sum of $\ell/3 = 111$ values ± 1). Then $\mathrm{PSD}_u(\ell/3) = |\gamma_0 + \gamma_1\zeta_3 + \gamma_2\zeta_3^2|^2 = \gamma_0^2 + \gamma_1^2 + \gamma_2^2 - \gamma_0\gamma_1 - \gamma_1\gamma_2 - \gamma_2\gamma_0$, the Eisenstein norm. Since the γ_r are odd and $1 + \zeta_3 + \zeta_3^2 = 0$, this value is divisible by 4: $\mathrm{PSD}_u(\ell/3) = 4(x^2 + xy + y^2)$ with $x = (\gamma_0 - \gamma_1)/2,\ y = (\gamma_1 - \gamma_2)/2 \in \mathbb{Z}$. With (1) this gives the necessary condition

$$\tfrac{\ell+1}{2} = 167 = L_u + L_v, \qquad L_u, L_v \text{ Loeschian (values of } x^2 + xy + y^2). \tag{2}$$

A short enumeration recovers exactly the Kotsireas–Koutschan spectrum [4]: 16 admissible values of $\mathrm{PSD}_u(\ell/3)$, all $\equiv 4 \pmod{12}$, forming 8 complementary Loeschian pairs summing to 167.

Ninth root (mod 9): the square-divisor refinement. Since $9 \mid 333$, one may also evaluate at a primitive 9th root $s = \ell/9$. With $c = c^{(9)}(u)$ (entries odd), the same parity argument gives $\mathrm{PSD}_u(\ell/9) = 4\,|E(\zeta_9)|^2$ for $E \in \mathbb{Z}[\zeta_9]$, a totally non-negative element of the cubic field $K = \mathbb{Q}(\zeta_9)^+ = \mathbb{Q}(\cos\frac{2\pi}{9})$, and (1) reads $167 = N_u + N_v$ in $\mathcal{O}_K$ with N_u, N_v totally non-negative. This is a genuine cubic-field condition, and no published filter uses a prime-power (rather than prime) frequency. One might hope it strengthens (2). It does not.

Proposition 3 (Square-divisor filter adds no pruning)**.** *For every length $\ell \in \{27, 45, 63\}$ (i.e. $\ell = 9m$ with $m \in \{3, 5, 7\}$), every* mod 3*-admissible state (γ, γ') (a pair of odd triples with row sums ± 1 and $\sum \gamma_r^2 + \sum \gamma_r'^2 = 4m + 2$) is the* 3*-compression of a valid length-*9 *compressed Legendre pair (c, c'). Hence the ninth-root condition is implied by the cube-root condition and provides no additional pruning of the spectrum.*

Verification. For each ℓ we enumerate all length-9 integer pairs (c, c') with odd entries in $[-m, m]$, row sums ± 1, and $\mathrm{PAF}_c(k) + \mathrm{PAF}_{c'}(k) = -2m$ ($k \neq 0$), $= 2\ell - 2m + 2$ ($k = 0$); we form the set of 3-compressions $(c^{(3)}, c'^{(3)})$ and check it equals the full set of mod 3-admissible states. The counts are: $\ell = 27$: 250,128 pairs, all 288 states realized; $\ell = 45$: 1,163,808 pairs, all 288 realized; $\ell = 63$: 9,385,956 pairs, all 756 realized. In every case the number of mod 3-admissible states that fail to lift is 0. $\square$

Remark 3. The case $\ell = 45$ has $(\ell + 1)/2 = 23$, prime and $\equiv 2 \pmod 3$ (the same arithmetic type as $167 = (333 + 1)/2$), so the negative result is tested on a length representative of 333, not merely on small toys. Structurally, the 9-compression space is enormous and is a weak filter; the cube-root norm already captures all spectral information available at the 3-part. We conclude that effort on LP(333) should not be directed at square-divisor spectral filters.

References

[1] D. Cati and D. V. Pasechnik, *Implementing Hadamard matrices in SageMath*, arXiv:2411.18897 (2024). [Records $668, 716, 892, 1132$ as the only orders ≤ 1208 of unknown status.]

[2] A. E. Brouwer and H. Van Maldeghem, *Strongly Regular Graphs*, Encyclopedia of Mathematics and its Applications **182**, Cambridge Univ. Press, 2022; and A. E. Brouwer, *Parameters of strongly regular graphs*, online tables (entry $(333, 166, 82, 83)$: open).

[3] R. J. Fletcher, M. Gysin and J. Seberry, *Application of the discrete Fourier transform to the search for generalised Legendre pairs and Hadamard matrices*, Australas. J. Combin. **23** (2001), 75–86.

[4] I. S. Kotsireas and C. Koutschan, *Legendre pairs of lengths $\ell \equiv 0 \pmod 3$*, J. Combin. Des. **29** (2021), 870–887; arXiv:2101.03116.

[5] N. A. Balonin and D. Ž. Đoković, *Three new Legendre pairs of lengths* $91, 93, 123$, arXiv:2010.02829 (2020).

[6] D. Ž. Đoković and I. S. Kotsireas, *Compression of periodic complementary sequences and applications*, Des. Codes Cryptogr. **74** (2015), 365–377; arXiv:1302.0571.

[7] J. J. Seidel, *A survey of two-graphs*, in *Colloquio Internazionale sulle Teorie Combinatorie* (Rome, 1973), 481–511, 1976.

[8] R. Mathon, *Symmetric conference matrices of order* $pq^2 + 1$, Canad. J. Math. **30** (1978), 321–331.

[9] C. Bright, I. Kotsireas and V. Ganesh, *When satisfiability solving meets symbolic computation*, Commun. ACM **65** (2022), no. 7, 64–72; and C. Bright, I. Kotsireas, D. Ž. Đoković, V. Ganesh, *A SAT+CAS method for enumerating Williamson matrices*, 2018.

[10] M. Behbahani and C. Lam, *Strongly regular graphs with non-trivial automorphisms*, Discrete Math. **311** (2011), 132–144.

[11] D. Crnković, M. Maksimović, B. G. Rodrigues and S. Rukavina, *Self-orthogonal codes from orbit matrices of strongly regular graphs*, Adv. Math. Commun. (2018).

[12] M. Maksimović, *Enumeration of regular two-graphs on up to* 50 *vertices via orbit matrices of prescribed automorphisms*, (2023).

[13] W. de Launey and D. L. Flannery, *Algebraic Design Theory*, Math. Surveys Monogr. **175**, Amer. Math. Soc., 2011.